

Shlok Jadhav

shlok.jadhav.07@gmail.com | (925) 389-4997 | linkedin.com/in/shlokjadhav | github.com/shlokrj | shlok.fyi

EDUCATION

University of Wisconsin-Madison

September 2025 - May 2028

Double Major in B.S. Computer Science & B.S. Data Science

GPA: 3.942, Dean's List: Fall 2025, Spring 2026

- **Relevant Coursework:** Comp Methods for Bio Model & Sim, Programming I, II, & III (Java), Data Sci Modeling I & II (R), Data Sci Programming I & II (Python), Computer Engineering, Machine Organization & Programming, Discrete Mathematics

PROFESSIONAL EXPERIENCE

TidalX AI

May 2026 - Present

Software Engineer Intern

Milpitas, CA

- Engineering high-throughput Python/scikit-learn pipelines for same-fish classification utilizing trajectory and image-pair data
- Performing feature engineering (SIFT/ORB, OpenCV) using keypoint/geometry features for biometric identity matching
- Developing HITL tools for image-pair curation and iterative optimization of Random Forest re-identification accuracy

Computational Biology & Machine Learning Lab

February 2026 - Present

Undergraduate Researcher

Madison, WI

- Building Python ML pipelines for birdsong analysis using spectrogram generation, audio preprocessing, and feature extraction
- Training encoder-decoder and LSTM-based models to learn compressed representations of bird vocalizations
- Applying PCA, UMAP, and topological data analysis to visualize learned embeddings and compare song structure

UW-Madison Department of Information Technology

January 2026 - Present

Advanced Help Desk Support Specialist

Madison, WI

- Providing technical support for software, accounts, and campus technology issues to on- and off-campus users via phone calls

Design Interactive

December 2025 - Present

Education Director

Madison, WI

- Leading weekly educational workshops for 50+ members on Figma, UI/UX design, and web development concepts
- Creating presentations to teach prototyping workflows, HTML/CSS, JavaScript, and frontend engineering fundamentals

Coding for SDG

September 2022 - June 2025

Secretary

San Francisco, CA

- Led 3 hackathons with 200+ participants and \$15,000+ in prizes across Coding for SDG 2023 and Hack the Planet 24 & 25

PROJECTS

Sweeper - AI Minesweeper Solver

Summer 2026

- Developed hybrid AI Minesweeper solver using symbolic constraints, frontier probability enumeration, and PyTorch CNN models
- Generated 20,000-board corpus and benchmarked CNNs, achieving 91.0% beginner win rate and +15.4-point expert improvement
- Engineered hybrid decision pipeline integrating symbolic reasoning, probabilistic inference, and calibrated CNN risk prediction

Retra - Diabetic Retinopathy Detection

Summer 2026

- Developed PyTorch/EfficientNet classification pipeline for automated diabetic retinopathy grading using APTOS fundus imagery
- Engineered retinal image preprocessing and augmentation workflows for improved model robustness and stronger generalization
- Implemented Grad-CAM activation maps and model evaluation with ROC curves, confusion matrices, and classification metrics

Cuey - Gesture-Controlled Media Controller

Spring 2026

- Developed MediaPipe/OpenCV pipeline for real-time hand landmark extraction and gesture classification for media control
- Engineered temporal smoothing, configurable gesture mappings, and low-latency cross-platform automation infrastructure

AWARDS & HONORARY MENTIONS

First Place Winner, NASA App Development Challenge

April 2024

- Developed a lunar visualization and pathfinding app for the Artemis Mission using the Ursina (Python) game engine
- Awarded the opportunity to present findings to NASA staff and faculty at the Johnson Space Center in Houston, TX
- Founded a student organization, NASA STEM, to guide high school students in competing for NASA STEM challenges

Second Place, JAXA Kibo Robot Programming Challenge

June 2024

- Programmed Astrobee controls, NASA's free-flying robot on the International Space Station via a Java-based Android application

Two-Time California State Finalist, Samsung Solve for Tomorrow

December 2023 & December 2024

- Designed wristbands to solve communication issues experienced by homeless youth and won a \$2,500 prize package in 2023
- Prototyped a device to diagnose diabetic retinopathy in underprivileged communities and won a \$2,500 prize package in 2024
- Competed as part of a student organization, XR EDU, serving as treasurer and managing club finances and internal operations

SKILLS & INTERESTS

Languages & Frameworks: Python, Java, R, C/C++, JavaScript/TypeScript, SQL, C#, React, Next.js, Node.js, FastAPI, HTML/CSS

AI/ML: PyTorch, Scikit-Learn, OpenCV, CNNs, Random Forests, NumPy, Pandas, SIFT/ORB, RANSAC, Constraint Propagation

Developer Tools: Git/GitHub, Linux/Unix, Bash, Docker, REST APIs, AWS, Microsoft Azure, Cloudflare Workers, Vercel, Jupyter